Part 1 — Database Setup

Create Tables

```
/* —— JOB_DEPARTMENT —— */
CREATE TABLE JOB_DEPARTMENT (
    job_ID        NUMBER PRIMARY KEY,
    job_dept      VARCHAR2(20)  NOT NULL,
    name          VARCHAR2(100) NOT NULL,
    description   VARCHAR2(200) NOT NULL,
    salary_range  VARCHAR2(30)  NOT NULL
);

/* —— SALARY_BONUS —— */
CREATE TABLE SALARY_BONUS (
    salary_ID  NUMBER PRIMARY KEY,
    amount     NUMBER(10,2) NOT NULL,
    annual     NUMBER(12,2) NOT NULL,
    bonus      NUMBER(10,2) NOT NULL,
    job_ID     NUMBER NOT NULL,
    CONSTRAINT chk_salary_amount CHECK (amount > 0),
    CONSTRAINT fk_salary_job FOREIGN KEY (job_ID) REFERENCES JOB_DEPARTMENT(job_ID)
);

/* —— EMPLOYEE —— */
CREATE TABLE EMPLOYEE (
    emp_ID       NUMBER PRIMARY KEY,
    fname        VARCHAR2(50)  NOT NULL,
    lname        VARCHAR2(50)  NOT NULL,
    gender       CHAR(1)       NOT NULL,
    age          NUMBER(3)     NOT NULL,
    contact_add  VARCHAR2(150) NOT NULL,
    emp_email    VARCHAR2(100) NOT NULL UNIQUE,
    emp_pass     VARCHAR2(100) NOT NULL,
    job_ID       NUMBER NOT NULL,
    salary_ID    NUMBER NOT NULL,
    CONSTRAINT chk_employee_gender CHECK (gender IN ('M','F')),
    CONSTRAINT fk_employee_job    FOREIGN KEY (job_ID)    REFERENCES
JOB_DEPARTMENT(job_ID),
    CONSTRAINT fk_employee_salary FOREIGN KEY (salary_ID) REFERENCES
SALARY_BONUS(salary_ID)
);

/* —— QUALIFICATION —— */
CREATE TABLE QUALIFICATION (
    qual_ID       NUMBER PRIMARY KEY,
    position      VARCHAR2(100) NOT NULL,
    requirements  VARCHAR2(200) NOT NULL,
    date_in       DATE NOT NULL,
    emp_ID        NUMBER NOT NULL,
    CONSTRAINT fk_qualification_employee FOREIGN KEY (emp_ID) REFERENCES
EMPLOYEE(emp_ID)
);

/* —— LEAVE —— */
CREATE TABLE LEAVE (
    leave_ID    NUMBER PRIMARY KEY,
    leave_date  DATE          NOT NULL,
    reason      VARCHAR2(100) NOT NULL,
    emp_ID      NUMBER NOT NULL,
    CONSTRAINT fk_leave_employee FOREIGN KEY (emp_ID) REFERENCES EMPLOYEE(emp_ID)
```

```
);

/* —— PAYROLL —— */
CREATE TABLE PAYROLL (
    payroll_ID    NUMBER PRIMARY KEY,
    pay_date      DATE          NOT NULL,
    report        VARCHAR2(200) NOT NULL,
    total_amount  NUMBER(10,2)  NOT NULL,
    emp_ID        NUMBER NOT NULL,
    job_ID        NUMBER NOT NULL,
    salary_ID     NUMBER NOT NULL,
    leave_ID      NUMBER,
    CONSTRAINT fk_payroll_employee FOREIGN KEY (emp_ID)      REFERENCES
EMPLOYEE(emp_ID),
    CONSTRAINT fk_payroll_job      FOREIGN KEY (job_ID)      REFERENCES
JOB_DEPARTMENT(job_ID),
    CONSTRAINT fk_payroll_salary   FOREIGN KEY (salary_ID)   REFERENCES
SALARY_BONUS(salary_ID),
    CONSTRAINT fk_payroll_leave    FOREIGN KEY (leave_ID)    REFERENCES
LEAVE(leave_ID)
);
```

Create Sequences

```
CREATE SEQUENCE JOB_DEPT_SEQ  START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE SALARY_SEQ    START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE EMP_SEQ       START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE QUAL_SEQ      START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE LEAVE_SEQ     START WITH 1 INCREMENT BY 1;
CREATE SEQUENCE PAYROLL_SEQ   START WITH 1 INCREMENT BY 1;
```

```
Sequence JOB_DEPT_SEQ created.


Sequence SALARY_SEQ created.


Sequence EMP_SEQ created.


Sequence QUAL_SEQ created.


Sequence LEAVE_SEQ created.


Sequence PAYROLL_SEQ created.
```

Part 2 — Data Insertion

JOB_DEPARTMENT

```
INSERT INTO JOB_DEPARTMENT VALUES (JOB_DEPT_SEQ.NEXTVAL,'ENG','Engineering','Software
and technical work','8000-10000');
INSERT INTO JOB_DEPARTMENT VALUES (JOB_DEPT_SEQ.NEXTVAL,'HR','Human
Resources','Employee services and recruitment','3000-4500');
INSERT INTO JOB_DEPARTMENT VALUES (JOB_DEPT_SEQ.NEXTVAL,'FIN','Finance','Accounting
and financial planning','5500-7000');
INSERT INTO JOB_DEPARTMENT VALUES (JOB_DEPT_SEQ.NEXTVAL,'MNT','Maintenance','Building
and equipment maintenance','2500-3500');
INSERT INTO JOB_DEPARTMENT VALUES (JOB_DEPT_SEQ.NEXTVAL,'OPS','Operations','Daily
business operations','4000-5000');
INSERT INTO JOB_DEPARTMENT VALUES (JOB_DEPT_SEQ.NEXTVAL,'LEG','Legal','Legal
support','4000-6000');
```

SALARY_BONUS

```
INSERT INTO SALARY_BONUS VALUES (SALARY_SEQ.NEXTVAL, 8000, 96000, 1000, 1);
INSERT INTO SALARY_BONUS VALUES (SALARY_SEQ.NEXTVAL, 3500, 42000,  500, 2);
INSERT INTO SALARY_BONUS VALUES (SALARY_SEQ.NEXTVAL, 6000, 72000,  800, 3);
INSERT INTO SALARY_BONUS VALUES (SALARY_SEQ.NEXTVAL, 2800, 33600,  300, 4);
INSERT INTO SALARY_BONUS VALUES (SALARY_SEQ.NEXTVAL, 4200, 50400,  600, 5);
```

EMPLOYEE

```
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Ahmed','Alhinai','M',28,'Muscat','ahmed@ems.com','pass123',1,1);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Fatma','Albalushi','F',31,'Muscat','fatma@ems.com','pass123',1,1);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Salim','Alshibli','M',35,'Sohar','salim@ems.com','pass123',2,2);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Mariam','Alrashdi','F',26,'Nizwa','mariam@ems.com','pass123',2,2);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Huda','Alkindi','F',39,'Muscat','huda@ems.com','pass123',3,3);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Said','Alhabsi','M',30,'Sur','said@ems.com','pass123',3,3);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Khalid','Almamari','M',42,'Rustaq','khalid@ems.com','pass123',4,4);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Noor','Alghafri','F',27,'Ibri','noor@ems.com','pass123',4,4);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Aisha','Almahrooqi','F',24,'Muscat','aisha@ems.com','pass123',5,5);
INSERT INTO EMPLOYEE VALUES
(EMP_SEQ.NEXTVAL,'Rashid','Albusaidi','M',40,'Barka','rashid@ems.com','pass123',5,5);
```

QUALIFICATION

```
INSERT INTO QUALIFICATION VALUES (QUAL_SEQ.NEXTVAL,'Software Engineer','Computer
Science Degree',DATE '2024-01-10',1);
INSERT INTO QUALIFICATION VALUES (QUAL_SEQ.NEXTVAL,'Software Engineer','Software
Engineering Degree',DATE '2024-02-15',2);
INSERT INTO QUALIFICATION VALUES (QUAL_SEQ.NEXTVAL,'HR Specialist','HR Diploma',DATE
'2023-03-01',3);
INSERT INTO QUALIFICATION VALUES (QUAL_SEQ.NEXTVAL,'Financial Analyst','Finance
Degree',DATE '2022-05-20',5);
INSERT INTO QUALIFICATION VALUES (QUAL_SEQ.NEXTVAL,'Technician','Maintenance
Diploma',DATE '2023-06-12',7);
```

LEAVE

```
INSERT INTO LEAVE VALUES (LEAVE_SEQ.NEXTVAL, DATE '2026-01-08', 'Sick Leave',   1);
INSERT INTO LEAVE VALUES (LEAVE_SEQ.NEXTVAL, DATE '2026-01-15', 'Annual Leave',  3);
INSERT INTO LEAVE VALUES (LEAVE_SEQ.NEXTVAL, DATE '2026-02-05', 'Sick Leave',    4);
INSERT INTO LEAVE VALUES (LEAVE_SEQ.NEXTVAL, DATE '2026-02-20', 'Family Leave',  6);
INSERT INTO LEAVE VALUES (LEAVE_SEQ.NEXTVAL, DATE '2026-03-01', 'Sick Leave',    7);
```

PAYROLL

```
INSERT INTO PAYROLL VALUES (PAYROLL_SEQ.NEXTVAL,DATE '2026-01-31','January Payroll',
9000,1,1,1,1);
INSERT INTO PAYROLL VALUES (PAYROLL_SEQ.NEXTVAL,DATE '2026-01-31','January Payroll',
9000,2,1,1,NULL);
INSERT INTO PAYROLL VALUES (PAYROLL_SEQ.NEXTVAL,DATE '2026-01-31','January Payroll',
4000,3,2,2,2);
INSERT INTO PAYROLL VALUES (PAYROLL_SEQ.NEXTVAL,DATE '2026-02-28','February
Payroll',4000,4,2,2,3);
INSERT INTO PAYROLL VALUES (PAYROLL_SEQ.NEXTVAL,DATE '2026-02-28','February
Payroll',6800,5,3,3,NULL);
INSERT INTO PAYROLL VALUES (PAYROLL_SEQ.NEXTVAL,DATE '2026-02-28','February
Payroll',6800,6,3,3,4);
INSERT INTO PAYROLL VALUES (PAYROLL_SEQ.NEXTVAL,DATE '2026-03-31','March Payroll',
3100,7,4,4,5);
INSERT INTO PAYROLL VALUES (PAYROLL_SEQ.NEXTVAL,DATE '2026-03-31','March Payroll',
4800,9,5,5,NULL);
COMMIT;
```

Verification Query

```
SELECT 'Departments'    AS table_name, COUNT(*) AS total FROM JOB_DEPARTMENT
UNION ALL SELECT 'Salary Bonus',   COUNT(*) FROM SALARY_BONUS
UNION ALL SELECT 'Employees',      COUNT(*) FROM EMPLOYEE
UNION ALL SELECT 'Qualifications', COUNT(*) FROM QUALIFICATION
UNION ALL SELECT 'Leaves',         COUNT(*) FROM LEAVE
UNION ALL SELECT 'Payroll',        COUNT(*) FROM PAYROLL;
```

Part 3 — SELECT Queries

Employees Aged Between 25 and 40

```
SELECT emp_ID, fname, lname, age
FROM   EMPLOYEE
WHERE  age BETWEEN 25 AND 40
ORDER BY lname ASC;
```

All Rows Fetched: 8 in 0.004 seconds

	EMP_ID	FNAME	LNAME	AGE
1	2	Fatma	Albalushi	31
2	10	Rashid	Albusaidi	40
3	8	Noor	Alghafri	27
4	6	Said	Alhabsi	30
5	1	Ahmed	Alhinai	28
6	5	Huda	Alkindi	39
7	4	Mariam	Alrashdi	26
8	3	Salim	Alshibli	35

Payroll Records Above 5,000

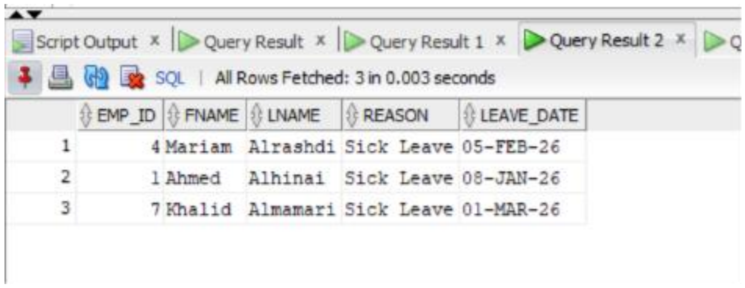
```
SELECT p.payroll_ID, e.fname, e.lname,
       jd.name AS department_name, p.total_amount
FROM   PAYROLL        p
JOIN   EMPLOYEE       e  ON p.emp_ID = e.emp_ID
JOIN   JOB_DEPARTMENT jd ON p.job_ID = jd.job_ID
WHERE  p.total_amount > 5000;
```

All Rows Fetched: 4 in 0.004 seconds

	PAYROLL_ID	FNAME	LNAME	DEPARTMENT_NAME	TOTAL_AMOUNT
1	1	Ahmed	Alhinai	Engineering	9000
2	2	Fatma	Albalushi	Engineering	9000
3	5	Huda	Alkindi	Finance	6800
4	6	Said	Alhabsi	Finance	6800

Employees Who Took Sick Leave

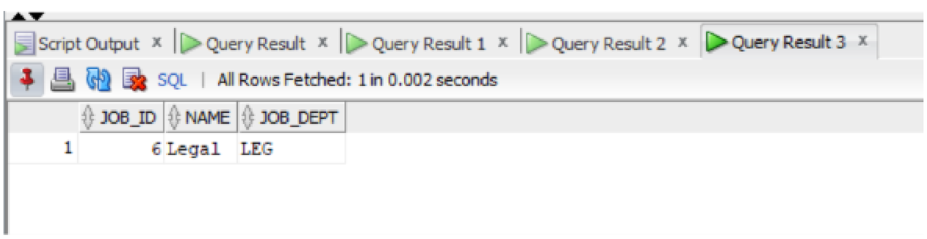
```
SELECT DISTINCT e.emp_ID, e.fname, e.lname,
       l.reason, l.leave_date
FROM   EMPLOYEE e
JOIN   LEAVE    l ON e.emp_ID = l.emp_ID
WHERE  LOWER(l.reason) LIKE '%sick%';
```



	EMP_ID	FNAME	LNAME	REASON	LEAVE_DATE
1	4	Mariam	Alrashdi	Sick Leave	05-FEB-26
2	1	Ahmed	Alhinai	Sick Leave	08-JAN-26
3	7	Khalid	Almamari	Sick Leave	01-MAR-26

Departments With No Employees

```
SELECT jd.job_ID, jd.name, jd.job_dept
FROM   JOB_DEPARTMENT jd
LEFT JOIN EMPLOYEE    e  ON jd.job_ID = e.job_ID
WHERE  e.emp_ID IS NULL;
```



	JOB_ID	NAME	JOB_DEPT
1	6	Legal	LEG